

Exercise 1 :

Example.java

```
package com.example;

public class Example {
    public int add(int a, int b) {
        return a + b;
    }
}
```

ExampleTest.java

```
package com.example;

import org.junit.Test;
import static org.junit.Assert.*;

public class ExampleTest {

    @Test
    public void testAdd() {
        Example example = new Example();
        int result = example.add(2, 3);
        assertEquals(5, result);
    }
}
```

Output :

```
Tests passed: 1 of 1 test - ExampleTest
Process finished with exit code 0
```

Exercise 3 :

Output :

```
Testing started at 5:43 PM ...
> Task :test

ExampleTest
  ✓ testAssertions() PASSED

BUILD SUCCESSFUL in 1s
1 test completed

Process finished with exit code 0
```

Exercise 4 :

Calculator.java

```
public class Calculator {

    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) {
        return a - b;
    }

    public int multiply(int a, int b) {
        return a * b;
    }

    public int divide(int a, int b) {
        if (b == 0) throw new ArithmeticException("Cannot divide by zero");
        return a / b;
    }
}
```

```
}  
}
```

CalculatorTest.java

```
import org.junit.After;  
import org.junit.Before;  
import org.junit.Test;  
import static org.junit.Assert.*;
```

```
public class CalculatorTest {
```

```
    private Calculator calculator;
```

```
    @Before
```

```
    public void setUp() {  
        calculator = new Calculator();  
        System.out.println("Setup done");  
    }
```

```
    @After
```

```
    public void tearDown() {  
        calculator = null;  
        System.out.println("Teardown done");  
    }
```

```
    @Test
```

```
    public void testAddition() {  
        int result = calculator.add(10, 5);  
        assertEquals(15, result);  
    }
```

```
    @Test
```

```
    public void testSubtraction() {  
        int result = calculator.subtract(20, 8);  
        assertEquals(12, result);  
    }
```

```
    @Test
```

```

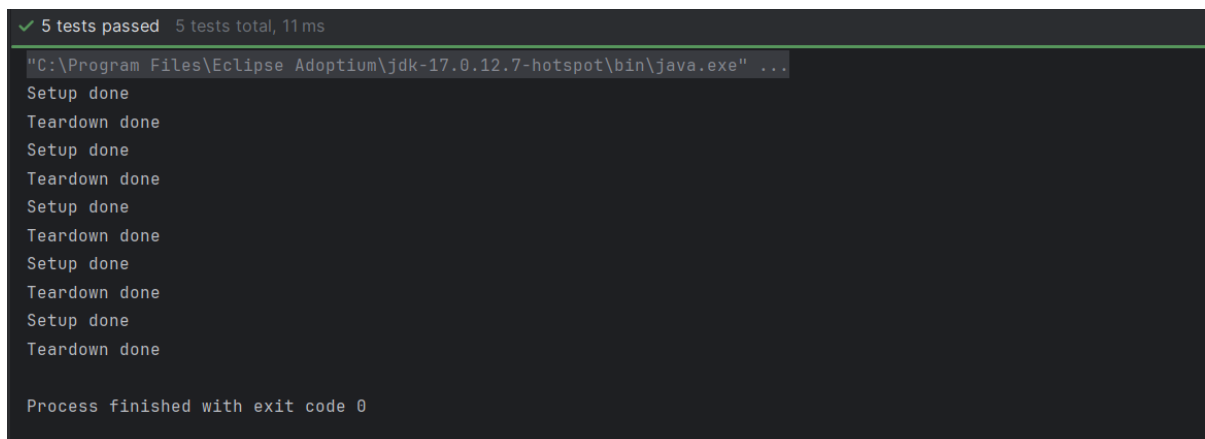
public void testMultiplication() {
    int result = calculator.multiply(4, 6);
    assertEquals(24, result);
}

@Test
public void testDivision() {
    int result = calculator.divide(20, 4);
    assertEquals(5, result);
}

@Test(expected = ArithmeticException.class)
public void testDivisionByZero() {
    calculator.divide(10, 0);
}
}

```

Output :



```

✓ 5 tests passed 5 tests total, 11 ms
"C:\Program Files\Eclipse Adoptium\jdk-17.0.12-hotspot\bin\java.exe" ...
Setup done
Teardown done
Setup done
Teardown done
Setup done
Teardown done
Setup done
Teardown done
Setup done
Teardown done
Process finished with exit code 0

```